# Introduction to Data Science and Systems

## Probabilities I – Introduction to probabilities

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# Today’s learning outcomes

- what probability is, and different philosophical interpretations of it
- what inverse and forward probability are
- what a random variable, distribution, probability mass/density function are
- what the empirical distribution is and how it is computed from data
- what expectation/expected value is
- the axioms of probability theory

# Example: the lost submarine

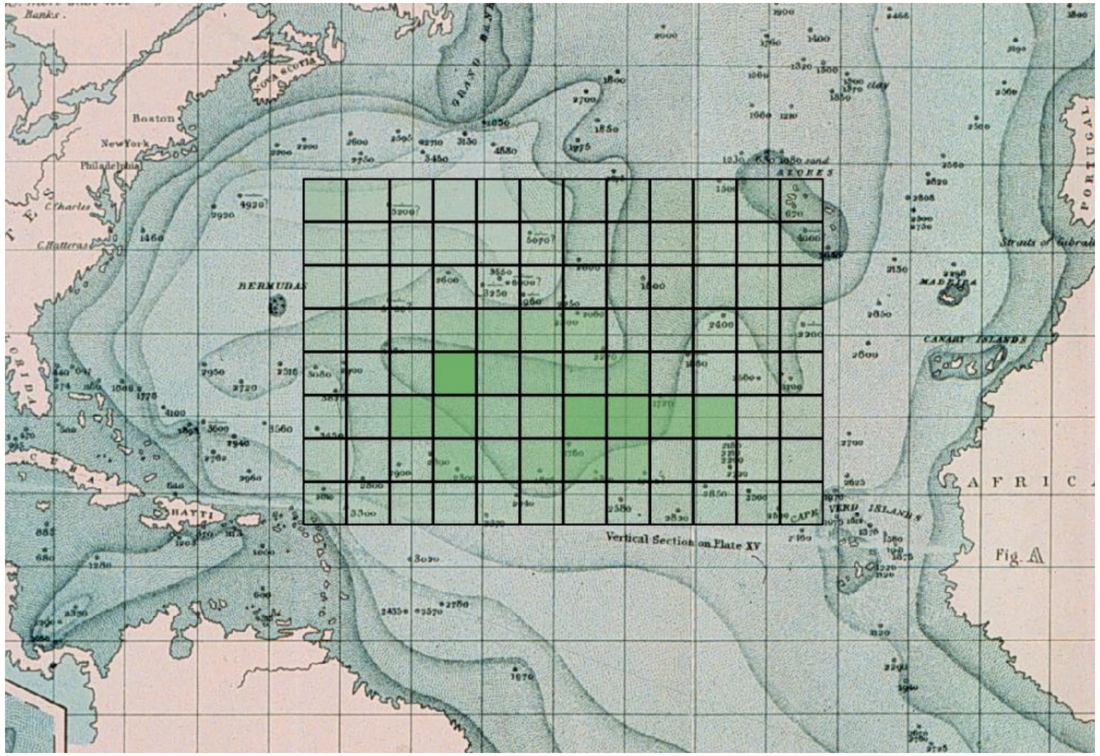

The *USS Scorpion* was a nuclear armed submarine that disappeared on 30th June 1968 somewhere in the Atlantic. [This was a year in which *four* submarines were inexplicably lost at sea -- **the 1968 submarine mystery**.]

The search for the submarine was one of the first times **probabilistic methods** were used for searching. The US Navy needed to find the submarine and recover it as fast as possible. Probabilistic methods allow us to reason *precisely* about things we are uncertain about -- like where the submarine is. Probability gives a concrete, computable representations for an uncertain world.

# Example: the lost submarine


Imagine dividing the possible search areas into grid squares: *what is the probability that the submarine lies within a specific square?*

A map like this might be produced, where squares are colored according to their probability:



# Part 1. Probability

- This section of the course is concerned with **stochastic elements**; the role of uncertainty, randomness and statistics in computation.
- The fundamental mathematical principles are drawn from **probability theory**, which gives us simple and powerful ways of manipulating uncertain values and let us do useful operations like inferring the most likely hypotheses given some observations.
- Probability theory is a simple, consistent, and effective way to manipulate uncertainty.

# What is probability?

- **Experiment** (or **trial**): An occurrence with an uncertain outcome.
  - *For example, losing a submarine -- the location of the submarine is now unknown.*
- **Outcome:** The result of an experiment; one particular state of the world.
  - *For example: the submarine is in ocean grid square [2,3].*
- **Sample Space:** The set of *all possible* outcomes for an experiment.
  - *For example, ocean grid squares {[0,0], [0,1], [0,2], [0,3], ..., [8,7], [8,8], [8,9], [9,9]}.*
- **Event:** A *subset* of possible outcomes with some common property.
  - *For example, the grid squares which are south of the Equator.*
- **Probability:** The probability of an event *with respect to a sample space* is the number of outcomes from the sample space that are in the event, divided by the total number of outcomes in the sample space. Since it is a ratio, probability will always be a real number between 0 (representing an impossible event) and 1 (representing a certain event).
  - *For example, the probability of the submarine being below the equator, or the probability of the submarine being in grid square [0,0] (in this case the event is just a single outcome).*

# Some definitions (cont'd)

- **Probability distribution:** A mapping of outcomes to probabilities that sum to 1. This is because an outcome must happen from a trial (with probability 1) so the sum of all possible outcomes together will be 1. A random variable has a probability distribution which maps each outcome to a probability.
  - *For example $P(X=x)$, the probability that the submarine is in a specific grid square x*.
- **Random variable:** A variable representing an unknown value, whose probability distribution we *do* know. The variable is associated outcomes of a trial
  - *For example, X is a random variable representing the location of the submarine.*
- **Probability density/mass function:** A function that *defines* a probability distribution by mapping each outcome to a probability $f_X(x), x \rightarrow \mathbb{R}$. This could be a continuous function over x (density) or discrete function over x (mass).
  - *For example $f_X(x)$ would be a probability mass function for the submarine, which maps each grid square to real number representing its probability.*

# Some definitions (cont'd)

- **Observation** An outcome that we have directly observed; i.e. data.
  - *For example, a submarine was found in grid square [0,5]*
- **Sample** An outcome that we have simulated according to a probability distribution. We say we have **drawn** a sample from a distribution.
  - *For example, if we believe that the submarine was distributed according to some pattern, generate possible concrete grid positions that follow this pattern.*
- **Expectation/expected value** The "average" value of a random variable.
  - *The submarine was on average in grid square [3.46, 2.19]*

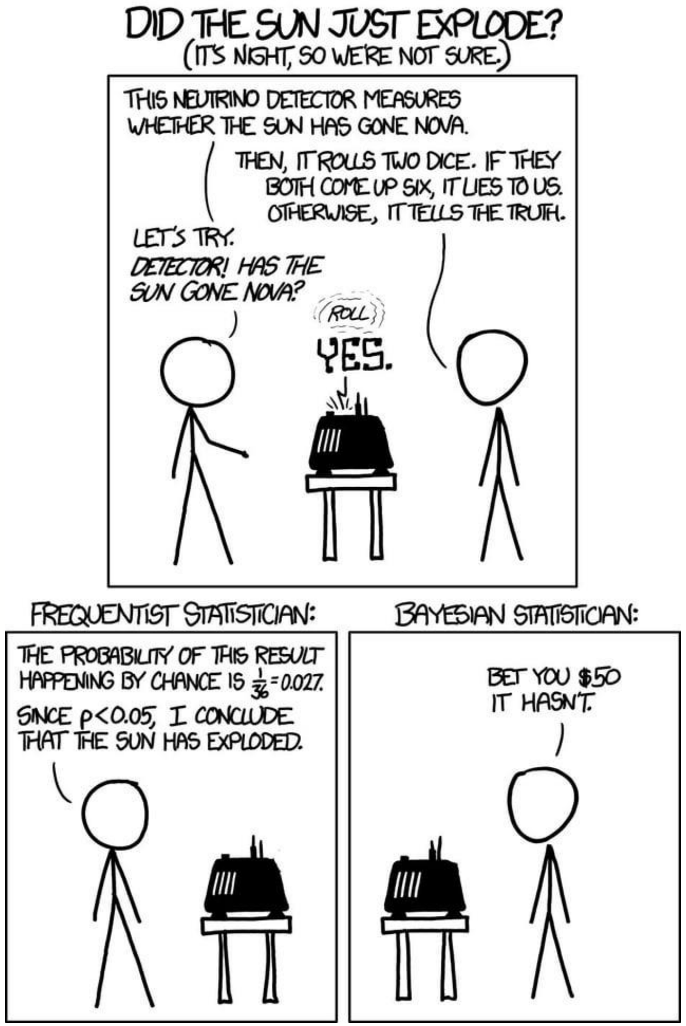
# In prose:

- A random variable $X$ *has* a probability distribution $P(X)$ which *assigns* probabilities $0 \leq P(X = x) \leq 1$ to *outcomes* $x$ which *belong* to a sample space $\mathbb{X}$.
- That probability distribution is *defined* by a probability density/mass function $fX(x)$ which assigns probabilities to outcomes such that the sum of probabilities over all outcomes is 1, $\sum_{x \in \mathbb{X}} f_X(x) = 1$.
- We can *observe* specific outcomes $x_i$ drawn from a distribution as a result of *trials*.
- We can *sample* (simulate) new outcomes $x'_j$ given a distribution P(X).
- Assuming outcomes have values we can *evaluate* the average expected value $\mathbb{E}[X]$ across infinitely many trials.

# Part 2. Philosophy of probability

There are two schools of thought regarding probability and its uses. We will be (largely) following the **Bayesian interpretation**, but its worth understanding what that entails.

- **Bayesian/Laplacian view**
- **Frequentist view**



# Bayesian/Laplacian view on probability

**Bayesians** treat probability as a **calculus of belief**;

- probabilities are measures of *degrees of belief*. $P(A)=0$ means a belief that event $A$ cannot be true and $P(A)=1$ is a belief that event $A$ is absolutely certain.
- In the Bayesian perspective, it makes sense to say "the probability it is raining outside is 0.3" (the probability quantifies our belief about the weather given the information we have). **Note that is not a statement that we believe that the weather is 0.3 rainy (whatever that means)**

# Bayesian/Laplacian view on probability (cont'd)

**Bayesians** allow for belief in states to be combined and manipulated via the rules of probability. The key process in Bayesian logic is *updating of beliefs*.

Given some:

- **prior** belief (it's Glasgow, it's not likely to be sunny) and some
- new **evidence** (there seems to be a bright reflection inside) we can
- update our belief to calculate the **posterior** -- our new probability that it is sunny outside.

Bayesian inference requires that we accept priors over events, i.e. that we must explicitly quantify our assumptions with probability distributions. It is an extension of logic to uncertain information.

# Bayesian/Laplacian view on probability (example)

For example, in the ***submarine search***:

- the ***prior*** might be that the submarine is probably in the south Atlantic (given the last radio broadcast received).
- ***Evidence*** might be the result of sonar surveys from search ships.
- After each survey, the ***posterior*** probability of the submarine being in the survey area could be updated. *This represents our belief about where the vessel might be.*

# Frequentist view of probability

There is an alternative school of thought that considers probabilities to *only* be the ***long-term behaviour of repeated events.***

- A **frequentist** does not accept phrases like "*what is the probability it is sunny just now?*" as there is no long-term behaviour involved (it is only "now" once). It does not make sense in this world view to talk about the probability of events that can only happen once.
- It *does* make sense in a frequentist view to ask things like "*what is the probability it will be sunny on any given day?*" since we can measure this event (sunny or not) for many different days.
- For example, frequentists would not assign a probability to the USS Scorpion being in a specific grid square; this is not an experiment that can be repeated.

# Objectivity and subjectivity

Frequentist versus Bayesian debates quickly enter philosophical territory. The diversity of viewpoints and depth of arguments cannot be done justice here.

- Very briefly, Bayesian probability theory is sometimes said to be **subjective** because it requires the specification of prior belief, whereas frequentist models of probability do not admit the concept of priors and thus is **objective**.
- An alternative view is that the Bayesian model explicitly encodes uncertain knowledge and states universal formal rules for manipulating that knowledge, as formal logic does for definite knowledge. Frequentist methods are objective in the sense that they make statements about universal truths (e.g. asymptotic behaviour), but they do not form a calculus of belief, and thus can't answer many questions of importance directly.

# Frequentist vs Bayesian (summary)

- **Bayesian**
  - Includes **priors**
  - Probability is a **degree of belief**
  - (Parameters of population considered to be random variables, data to be known)
- **Frequentist**
  - No **priors**
  - Probability is the *long-term frequency of events*
  - (Parameters of population assumed to be fixed, data to be random)

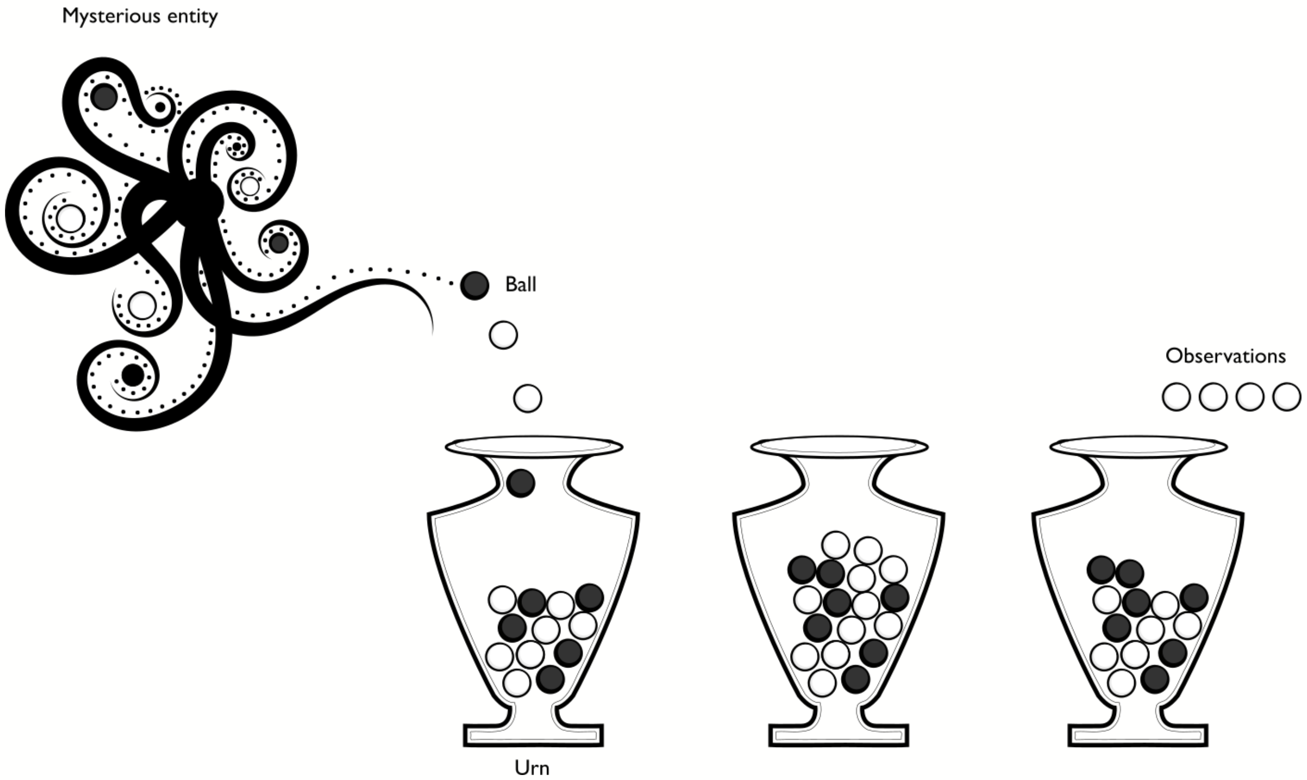
# Superiority of probabilistic models

Regardless of the philosophical model you subscribe to, there is one thing you can be sure of: *probability is the best*.

There are other models of uncertainty than probability theory that are sometimes used. However, all other representations of uncertainty are *strictly inferior* to probabilistic methods *in the sense that* a person, agent, computer placing "bets" on future events using probabilistic models has the best possible return out of all decision systems when there is uncertainty.

*Any theory with as good a gambling outcome as would be achieved using probability theory is equivalent to probability theory.*

# Generative models




Consider an urn, into which a number of balls have been poured (by some mysterious entity, say). Each ball can be either black or white.

You pull out four random balls from the urn and observe their colour. You get four white balls.

# Generative models

There are lots of questions you can ask now:

**What is the probability that the next ball that is drawn will be white?**

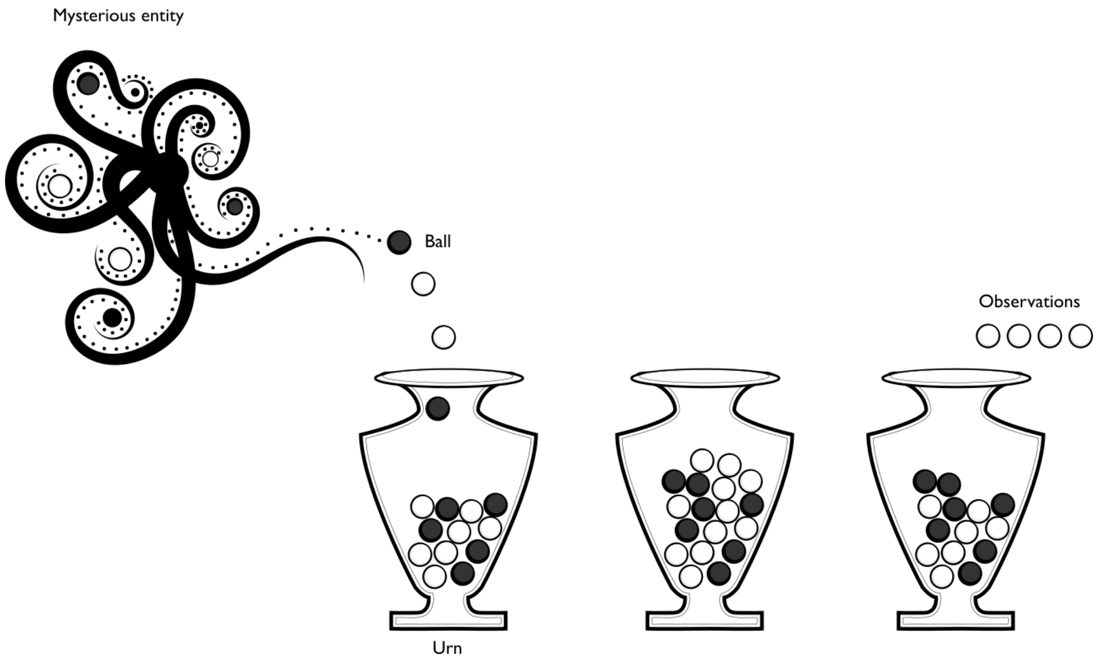
- This is a **forward probability** question. It asks questions related to the distribution of the observations.

**What is the distribution of white and black balls in the urn?**

- This is an **inverse probability** question. It asks questions related to unobserved variables that govern the process that generated the observations.

**Who is the mysterious entity?**

- This is an unknowable question. The observations we make cannot resolve this question.



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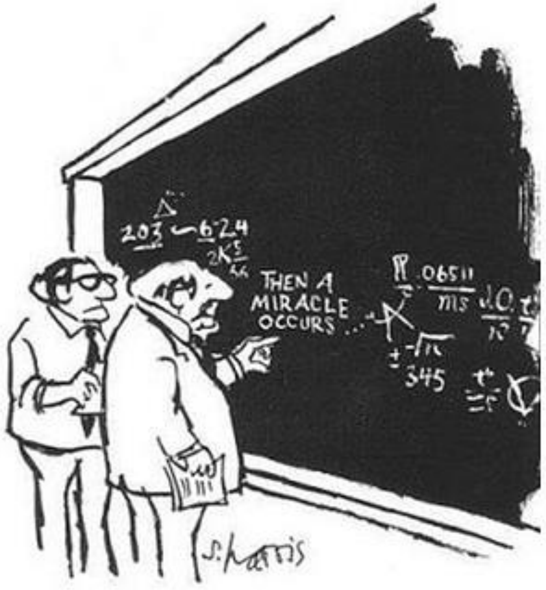
# Part 4. A formal basis for probability theory




"I THINK YOU SHOULD BE MORE EXPLICIT HERE IN STEP TWO."

Sidney Harris (1977), The New Yorker

# Axioms of probability

- **Boundedness**

$$0 \leq P(A) \leq 1$$

  all possible events $A$ -- probabilities are 0, or positive and less than 1.

- **Unitarity**

$$\sum_x P(x) = 1$$

  for the complete set of possible **outcomes** (not events!) $x \in \sigma$ in a sample space $\sigma$ -- something always happens.

- **Sum rule**

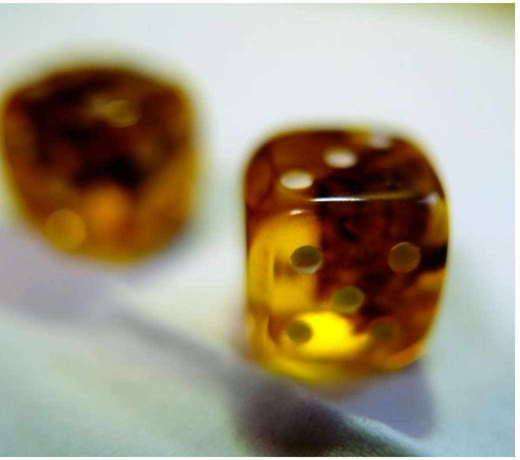
$$P(A \vee B) = P(A) + P(B) - P(A \wedge B),$$

  i.e. the probability of either event $A$ or $B$ happening is the sum of the independent probabilities minus the probability of both happening. (notation note: $\vee$ means "or" and $\wedge$ means "and")

- **Conditional probability** The conditional probability $P(A|B)$ is defined to be the probability that event $A$ will happen *given that we already know $B$ to have happened*.

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$

# Random variables and distributions

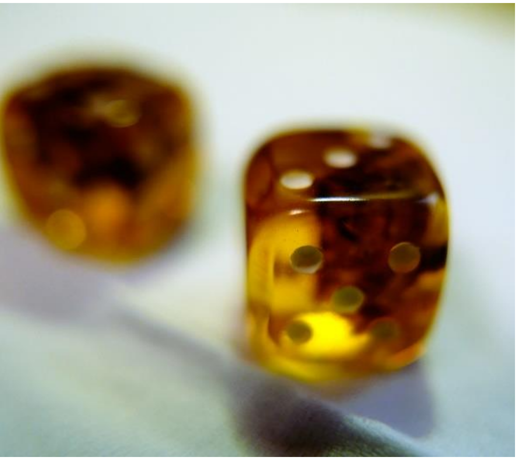


A **random variable** is a variable that can take on different values, but we do not know what value it has; i.e. one that is "unassigned".

However, we have *some knowledge* which captures the possible states the variable could take on, and their corresponding probabilities.

Probability theory allows us to manipulate random variables without having to assign them a specific value.

# Random variables and distributions

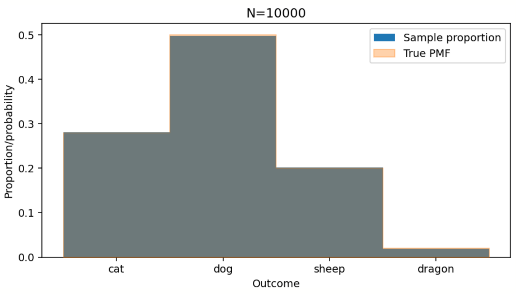


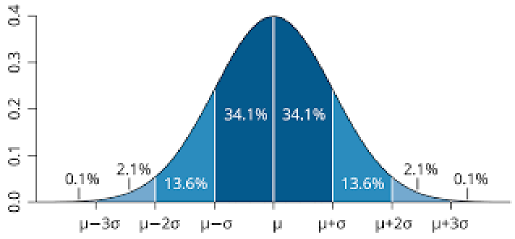
A random variable is written with a capital letter, like $X$.

A random variable might represent:

- the outcome of dice throw (discrete);
- whether or not it is raining outside (discrete: binary);
- the latitude of the USS Scorpion (continuous);
- the height of person we haven't met yet (continuous).

# Distributions




A **probability distribution** defines how likely different states of a random variable are.

We can see $X$ as the *experiment* and $x$ as the *outcome*, with a function mapping every possible outcome to a probability.

We write $P(X=x)$ (note the case!), and use the shorthand notations:

- $P(X=x)$ the probability of random variable $X$ taking on value $x$
- $P(X)$ shorthand for probability of $X=x$
- $P(x)$ shorthand for probability of specific value $X=x$

# Discrete and continuous

Random variables can be continuous (e.g. the height of a person) or discrete (the value showing on the face of a dice).

- **Discrete variables:** The distribution of a discrete random variable is described with a **probability mass function** (PMF) which gives each outcome a specific value; imagine a Python dictionary mapping outcomes to probabilities. The PMF is usually written $f_X(x)$, where $P(X=x)=f_X(x)$.
- **Continuous variables:** A continuous variable has a **probability density function** (PDF) which specifies the spread of the probability over outcomes as a *continuous function* $f_X(x)$.
  - It is **not** the case that $P(X=x) = f_X(x)$ for PDFs.

# Integration to unity

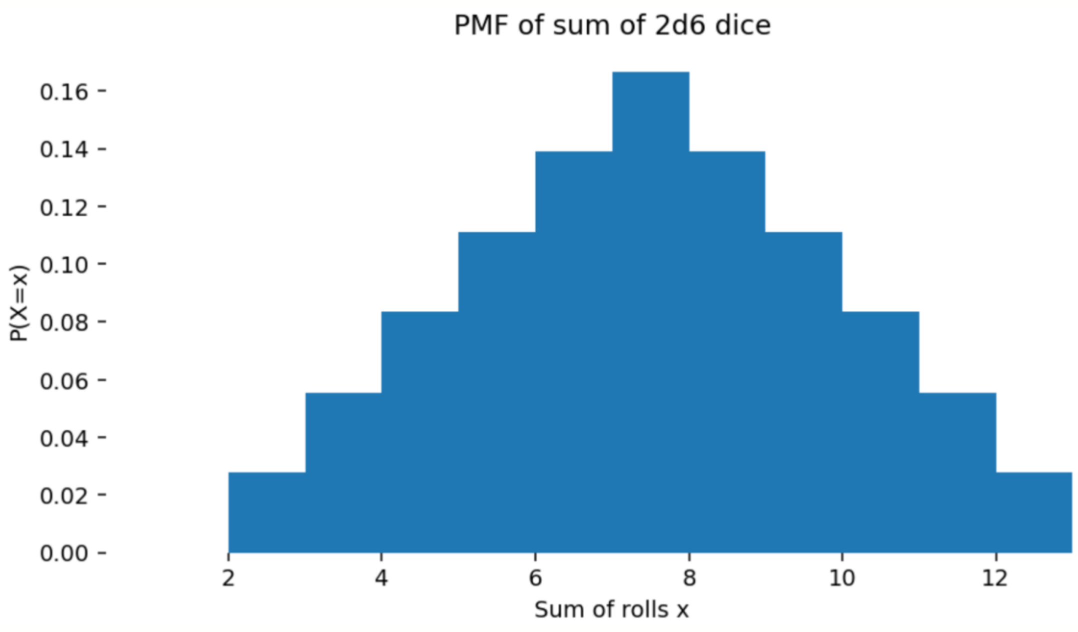
A probability mass function or probability density function *must* sum/integrate to exactly 1, as the random variable under consideration must take on *some* value; this is a consequence of unitarity.

*Every repetition of an experiment has exactly one outcome.*

$$\sum_i f_X(x_i) = 1 \quad \text{for PMFs of discrete RVs}$$

$$\int_x f_X(x)\, dx = 1 \quad \text{for PDFs of continuous RVs}$$

# PMF example: Sum of dice rolls




A very simple discrete PMF is the expected value of the sum of two six-faced dice.

$P(X=x)=f_X(x)$ takes on values for each possible outcome $x \in \{2,3,4,5,6,7,8,9,10,11,12\}$

# 5. Expectation

- The expectation is the "average" of a random variable. it represents what we'd "expect to happen"; the most likely overall "score".
- It can be thought of as a *weighted sum* of all the possible outcomes of an experiment, where each outcome is weighted by the probability of that outcome occurring.
- For example, in the "pair of dice" scenario, we can compute the expected value of the number of "dots" showing in total after a roll. We compute the probability of each number of dots showing and multiply by that number of dots, and summing the result. This is the expected number of dots showing, on average, or the expectation.

# Expectation – definition

If a random variable takes on numerical values, then we can define the **expectation** or **expected value** of a random variable $\mathbb{E}[X]$ as:

$$\mathbb{E}[X] = \int_x x\ f_X(x) dx$$

For a discrete random variable with probability mass function $P(X = x) = f_X(x)$, we would write this as a summation:

$$\mathbb{E}[X] = \sum_x f_X(x) x$$

If there are only a finite number of possibilities, then this is:

$$\mathbb{E}[X] = P(X = x_1)x_1 + P(X = x_2)x_2 + \cdots + P(X = x_n)x_n$$

# Expectation: Dice example

[30]:
```
from collections import Counter
dice_values =  [i+j for i in range(1,7) for j in range(1,7)]

# count the occurence of each possible value
counts = Counter(dice_values)
total_possibilities = len(dice_values)

pmf = np.array([counts[i]/total_possibilities for i in counts])
print("PMF of two dice:", pmf)
values = np.array([i for i in counts])
print("Face value of two dice:", values)

# expectation is each value weighted by probability.
expected_value = np.sum(pmf * values)
print("Expected value of sum of two dice: %.2f" % expected_value)
```

```
PMF of two dice: [0.02777778 0.05555556 0.08333333 0.11111111 0.13888889 0.16666667
 0.13888889 0.11111111 0.08333333 0.05555556 0.02777778]
Face value of two dice: [ 2  3  4  5  6  7  8  9 10 11 12]
Expected value of sum of two dice: 7.00
```

# Expectation

This is an intuitive property.

Imagine you meet a street hustler, who asks you to play the two dice game.

*He offers you a chance to buy in for £8; you win as many pounds as the show on the top faces of the dice after throwing them.*

Is this a fair game?

No. The expected return is only £7, and you have to put in £8 to play, so the expected result is a £1 loss (sometimes stated as "negative expected value" or -ve EV). If it was £7 to buy in, the game would be fair, in the sense that you and the hustler would not transfer money on average.

# Expectation and mean

Expectation corresponds to the idea of a **mean** or **average** result.

The expected value of a random variable is the **true average** of the value of all outcomes that would be observed if we ran the experiment an infinite number of times. This is the **population mean** -- the mean of the whole, possibly infinite, population of a random variable.

Many important properties of random variables can be defined in terms of expectation.

- The mean of a random variable $X$ is just $E[X]$. It is a measure of **central tendency**.
- The variance of a random variable $X$ is $var(X)=E[(X-E[X])2]$. It is a measure of **spread**.

# Expectation of functions of X

We can apply functions to random variables, for example, the square of a random variable.

The expectation of any function $g(X)$ of a continuous random variable $X$ is defined as:

$$\mathbb{E}[g(X)] = \int_x f_X(x)g(x)dx$$

or

$$\mathbb{E}[g(X)] = \sum_x f_X(x)g(x)$$

for a discrete random variable.

Be very careful:

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

# Expectation of functions of X

- For example, we can compute simple expectations like these:

$$\mathbb{E}[2X^2] = \sum_x f_X(x) 2x^2 dx$$

$$\mathbb{E}[\sin(X)] = \sum_x f_X(x) \sin(x) dx$$

```
[33]: sqr_expected_value = np.sum(pmf * (np.arange(2,13))**2)
      print("Expected value of square of sum of two dice: %.2f" % sqr_expected_value)

Expected value of square of sum of two dice: 54.83
```

# Expected values and decisions

Expected values are essential in making **rational decisions**, the central problem of **decision theory**. They combine scores (or **utility**) with uncertainty (**probability**).

The expected value gives us a way of deciding, for example, how much it would be worth paying to play a dice game.

- If the units were pounds, we'd break even if we paid £2.33 to play the game and make a profit (on average) if we paid £2.00 to play the game.
- The expected average profit is just the stake we pay to play each game minus the expected value of one round of the game:

$$\mathbb{E}[\text{game}] = \text{stake} - \mathbb{E}[X]$$

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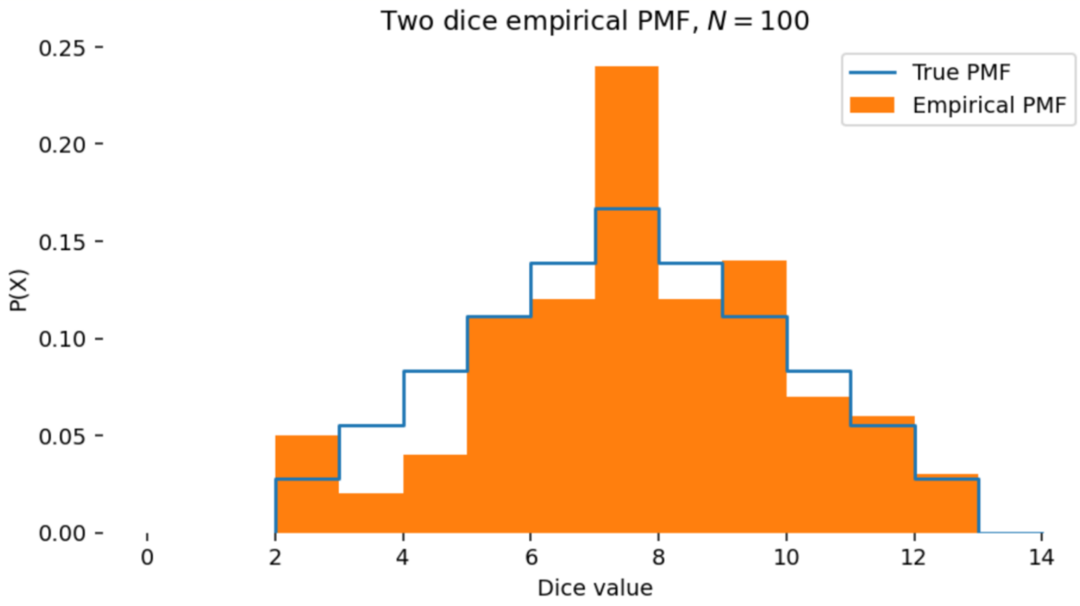
# 6. Samples and sampling

**Samples** are observed outcomes of an experiment; we will use the term **observations** synonymously, though samples usually refer to simulations and observations to concrete real data.

- We can **sample** from a distribution; this means simulating outcomes according to the probability distribution of those variables.
- We can also **observe** data which comes from an external source that we might believe is generated by some probability distribution.

For example, we can sample from the sum of dice PMF by rolling two dice and summing the result. This is a *sample* or a *draw* from this distribution. For discrete random variables, this is easy: we simply produce samples by drawing each outcome according to its probability.

# The empirical distribution




For discrete data, we can estimate the PMF that might be generating **observations** by counting each outcome seen divided by the total number of trials. This is called the **empirical distribution**.

This can be thought of as the **normalized histogram** of counts of occurrences of outcomes.

# Computing the empirical distribution

For discrete random variables, we can always compute the empirical distribution from a series of observations.

For example, from the counts of a specific word in a *corpus* of text (e.g. in every newspaper article printed in 1994). We just count the number of times each word is seen and divide by the total number of words.

$$P(X = x) = \frac{n_x}{N},$$

where $n_x$ is the number of time outcome $x$ was observed, and $N$ is the total number of trials.

Note that the empirical distribution is a distribution which *approximates* an unknown true distribution. When N is large it approximates the true PMF assuming the samples are drawn in an umbiased way.

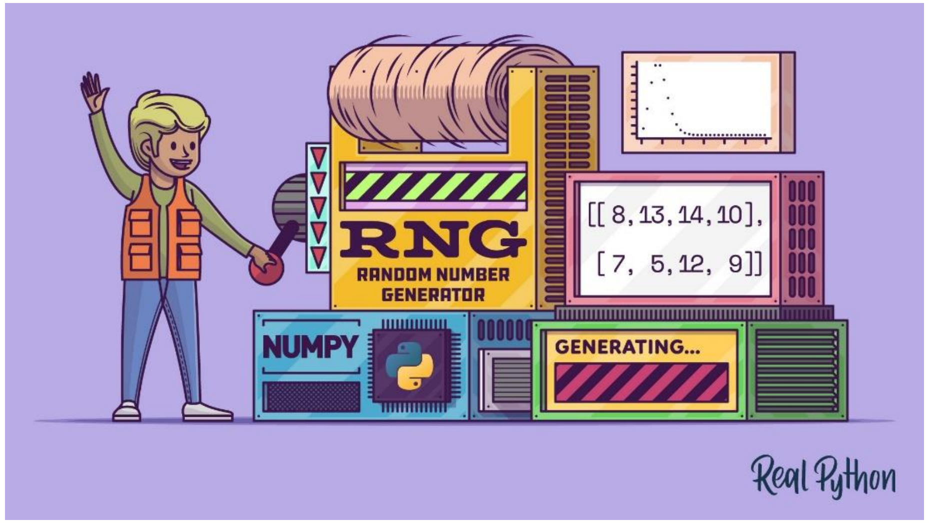
However, this approach does not work usefully for continuous random variables, since we will only ever see each observed value once (think about why!).

# Random numbers

There are algorithms which can generate continuous random numbers which are uniformly distributed in an interval, such as from 0.0 to 1.0.

These are actually pseudo-random numbers in practice (computers are deterministic!) but approximate the statistical properties of true random sequences.

We must be careful: computers generate pseudo-random floating-point numbers; and not true random real numbers. While this makes little difference much of the time, they are quite different things.




https://realpython.com/numpy-random-number-generator/

```
[45]: print(np.random.uniform(0,1)) # 0.0 - 1.0
      print(np.random.uniform(-np.pi, np.pi, (10,))) # -pi to pi, 10 elements
      print(np.random.uniform(1,100, (2,2))) # 1 to 100, 2x2 matrix
```

```
0.5490503399874326
[ 2.52262638  2.76430257 -0.08731961 -1.71925146  1.94226437  2.83232596
 -1.36734717  0.59294085  0.06219524 -2.62729178]
[[78.99383669 51.74073217]
 [89.35093217 72.99425164]]
```

# Uniform sampling

A **uniformly distributed** number has equal probability of taking on any value in its interval and zero probability everywhere else.

Although this is sampling from a continuous PDF, it is the key building block in sampling from arbitrary PMFs.

A uniform distribution is notated $X \sim U(a,b)$, meaning $X$ is random variable which may take on values between $a$ and $b$, with equal possibility of any number in that interval.

The symbol $\sim$ is read "distributed as", i.e. "X is distributed as a uniform distribution in the interval [a,b]".

Note that in practice these are not uniform across the reals in a given interval if we are using floating point, because we can only ever sample valid floating point values.

# Discrete sampling

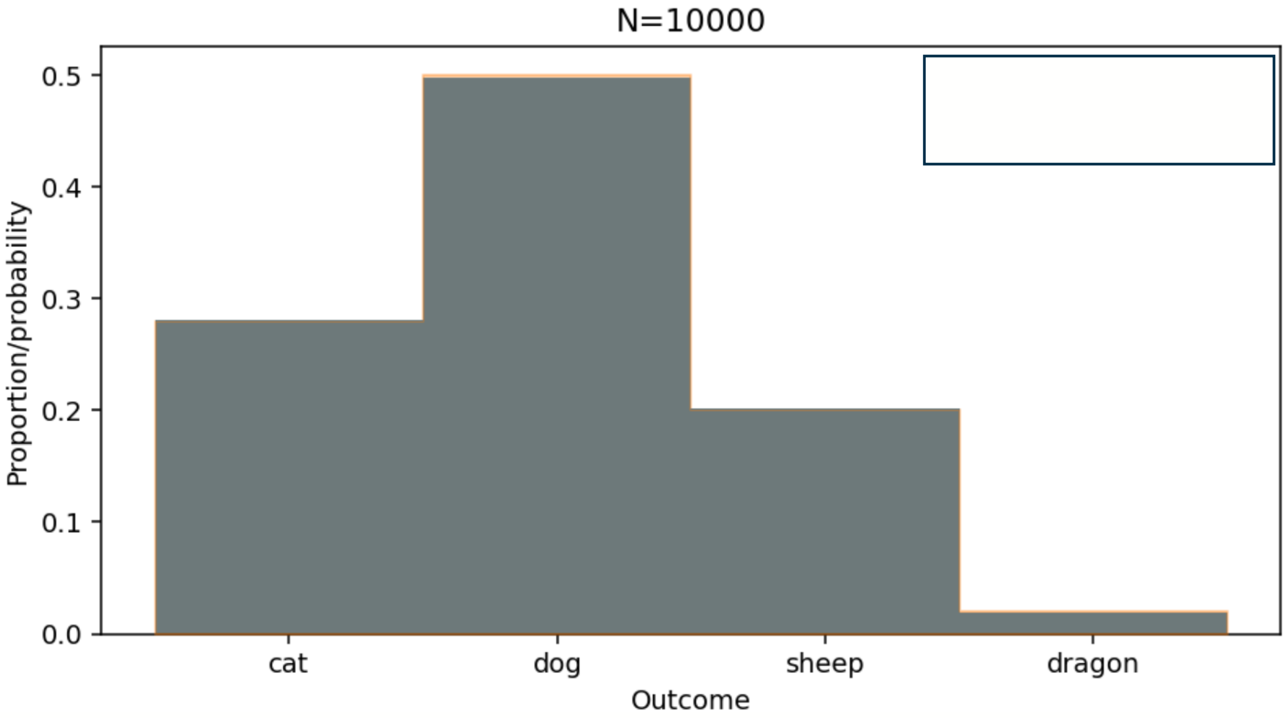
For a discrete probability mass function, we can sample outcomes according to *any arbitrary PMF* by partitioning the unit interval.

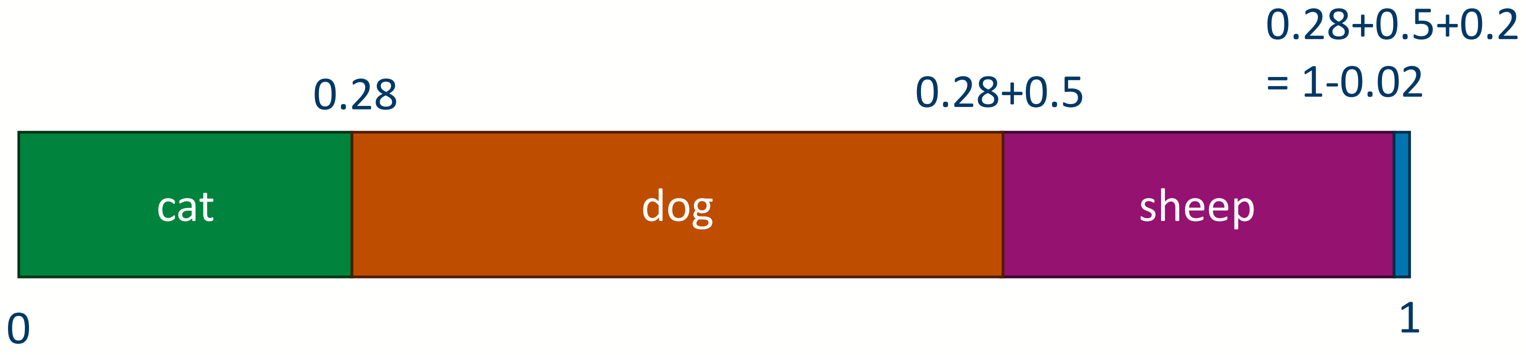
**Algorithm:**

1. choose any arbitrary ordering for the outcomes $x_1,x_2,\ldots$
2. assign each outcome a "bin" which is a portion of the interval [0,1] equal to its probability, so that the interval is divided into consecutive non-overlapping regions $[P(x_1) \rightarrow P(x_1)+P(x_2)\ ,\ P(x_1)+P(x_2) \rightarrow P(x_1)+P(x_2)+P(x_3)\ ,\ \ldots]$
3. draw a uniform sample in the range [0,1]
4. whichever "outcome bin" it lands in is the sample to draw

By the definition of a PMF, the sum of all the probabilities will be 1.0, so it will fill the interval [0,1] perfectly with no gaps.

# Discrete sampling example



# Example: Romeo & Juliet

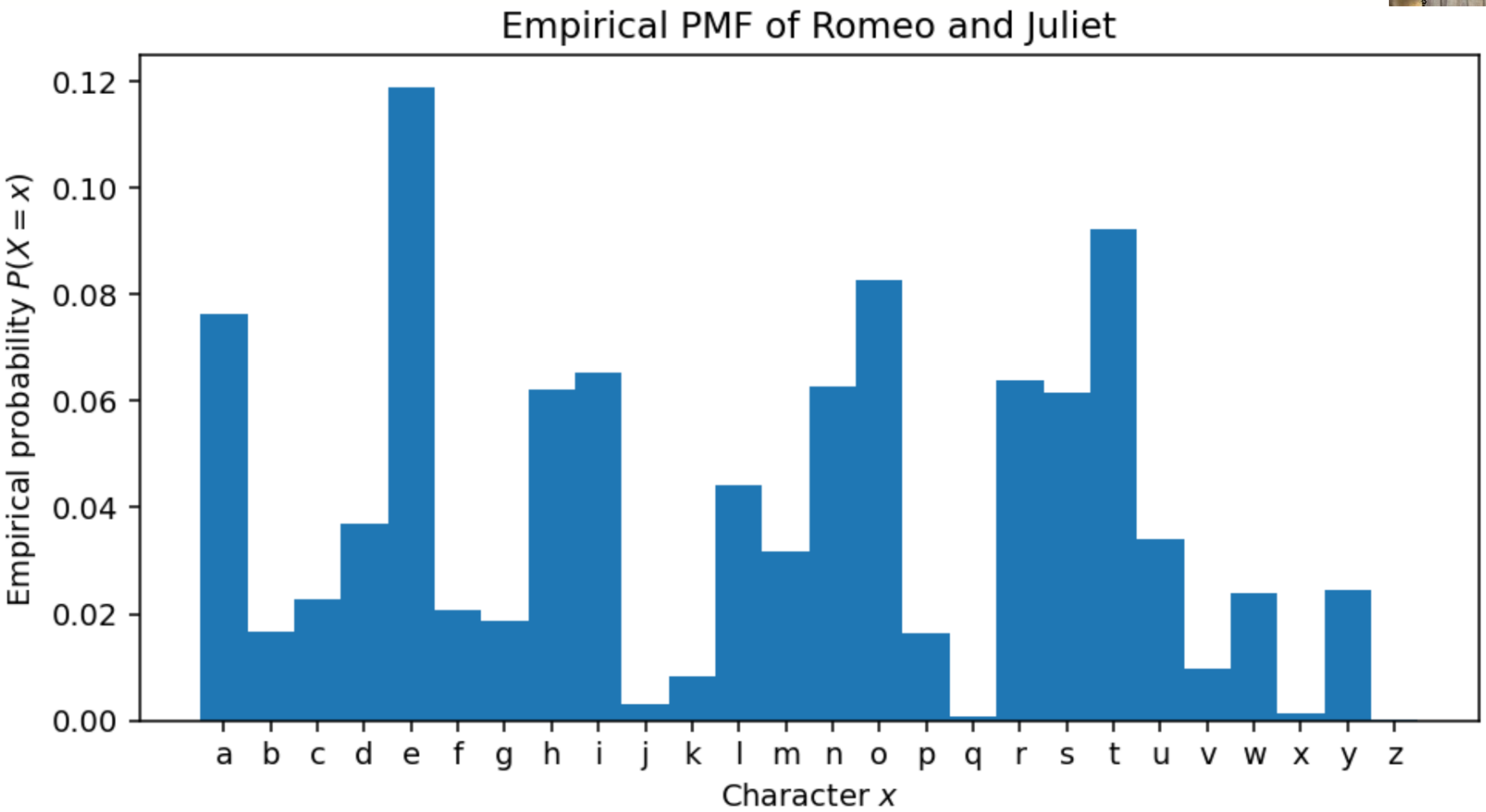
The code in the lecture notes loads a text file (in this case, Romeo and Juliet), and converts into a vector of numerical codes. It keeps only letters and spaces and converts everything to lowercase.

*Rom. Give me a torch. I am not for this ambling.*
*Being but heavy, I will bear the light.*

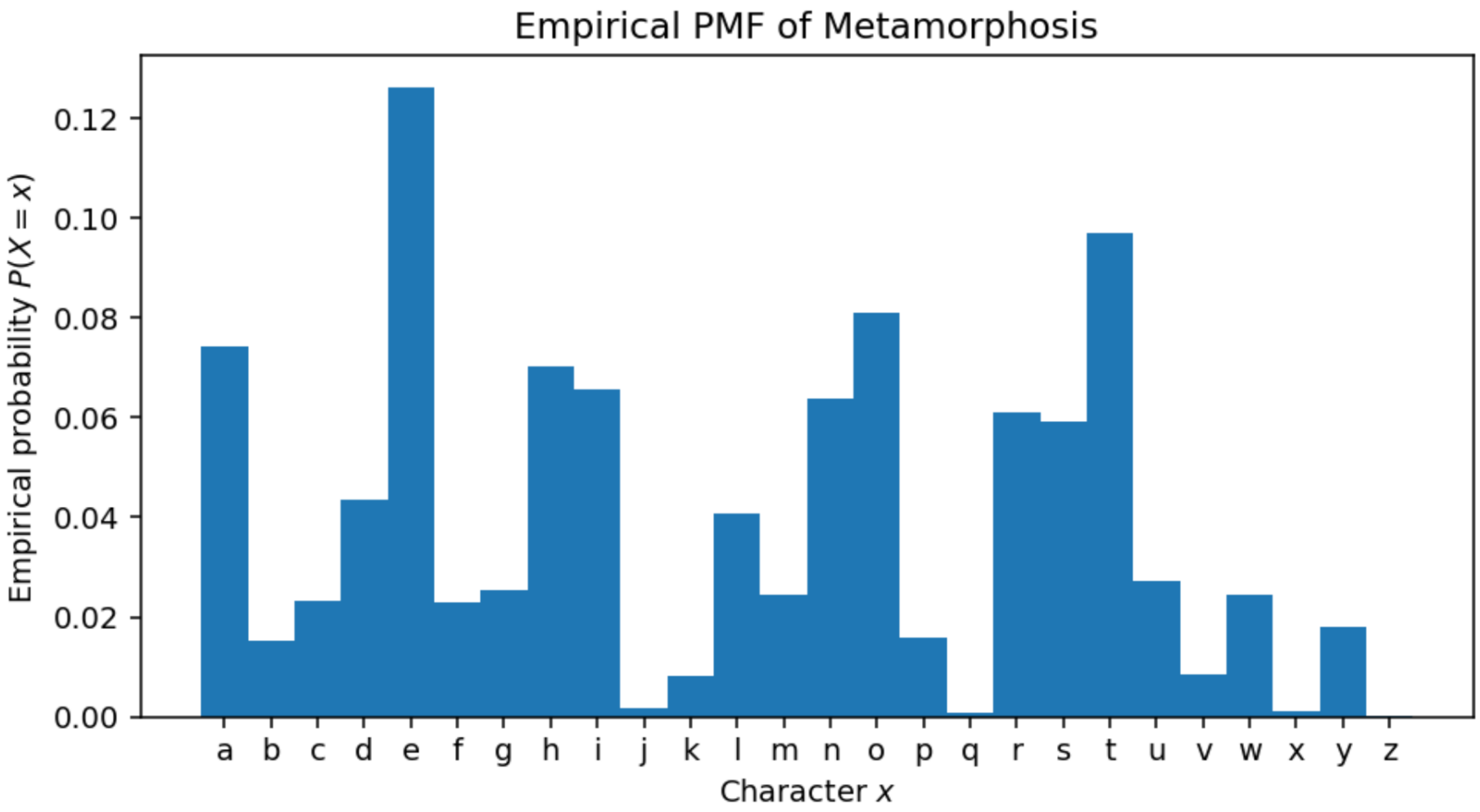
*Mer. Nay, gentle Romeo, we must have you dance.*

*Rom. Not I, believe me. You have dancing shoes*
*With nimble soles; I have a soul of lead*
*So stakes me to the ground I cannot move.*

# Letters PMF in Romeo & Juliet



# Letters PMF in Metamorphosis




# Judging a book (without its cover!)

The results are similar (both are English texts), but there are some subtle differences; there slightly more q characters in Kafka compared to Shakespeare; the frequencies of h and i are reversed.

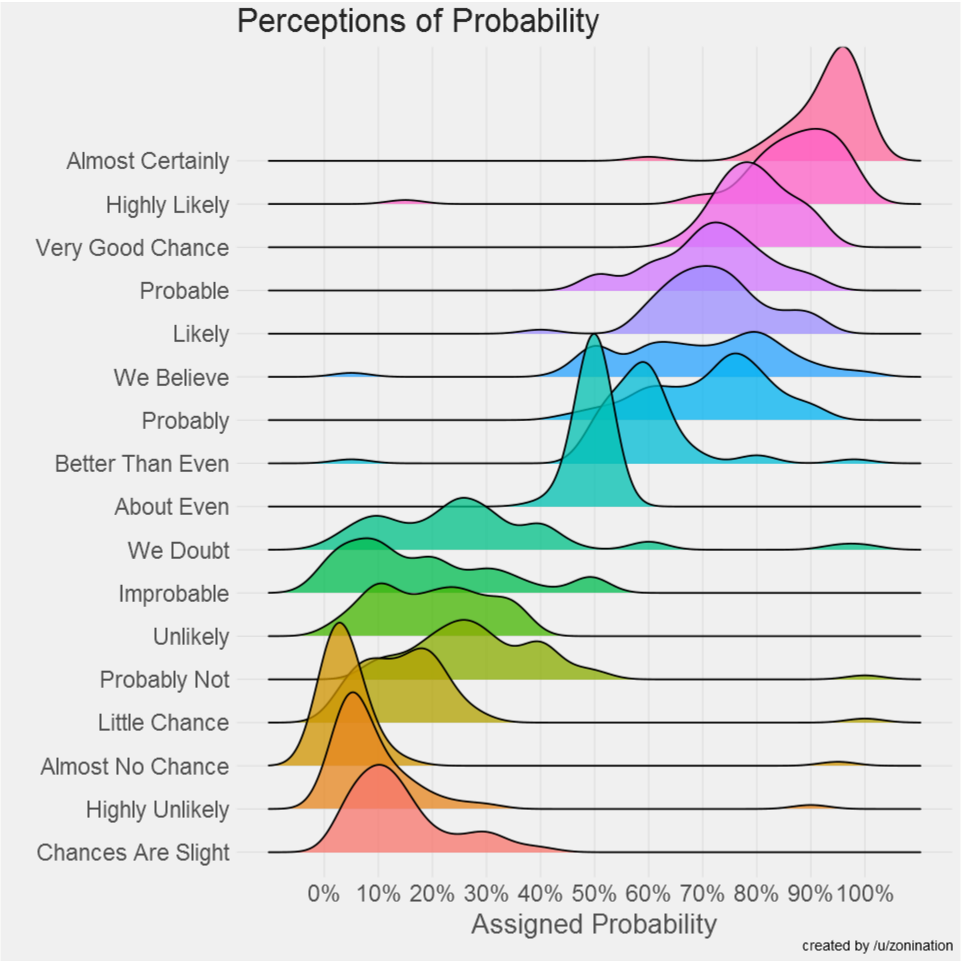
As always, given a PMF, we can draw samples from it with our standard procedure, though the results aren't very exciting:

```
[60]: dict_pmf = {outcome:p for outcome,p in zip(alphabet, rj_pmf)}
      "".join(discrete_sample(dict_pmf, 140))
```

```
[60]: 'flfsuagsgdhgumierotoaecnoobhftlrkhbudmgrgslseadmtiigmwhnmeteetbcrihgenietesiioasktorntmpoeahamnyrhfuhgqteirltspamyeeitaagiaeaebntnethbslbbam'
```

# 7. Perceptions of probabilities

- Raw probabilities (e.g. P(X=x)=0.9999) are not always intuitive.
- The graph below shows the probability that a sample of people imagined, given specific verbal cues.
- What about 1 in a million events? What about the probability of the sun rising tomorrow?
- Even if people had responded accurately, the linear visualisation from 0% to 100% makes it very hard to see extreme values.



# Odds & log odds

- The **odds** of an event with probability p is defined by:

$$\text{odds} = \frac{p}{1-p}$$

The odds are a more useful unit for discussing unlikely scenarios (odds of 1:999, or *999:1 against*, is easier to understand than p=0.001).

# Odds: example

```
[25]: p = 0.18

      def odds(p):
          return (p) / (1-p)

      print("p %.4f => odds %.2f:1, odds against %.2f:1" %(p,odds(p), odds(1-p)))
```

```
p 0.1800 => odds 0.22:1, odds against 4.56:1
```

# Log odds

**Log-odds** or **logit** are particularly useful for very unlikely scenarios:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

The logit scales proportionally to the number of zeros in the numerator of the odds.

# Log odds example

[26]:

```python
def logit(p):
    return np.log((p)/(1-p))

def print_logit(p):
    print("p {p: 12.8f} odds {odds: 12.1f}   log odds {log_odds:12.4f}".format(p=p, odds=odds(p), log_odds=logit(p)))

print_logit(0.1)
print_logit(0.01)
print_logit(0.001)
print_logit(0.00000003)
```

```
p   0.10000000 odds          0.1   log odds      -2.1972
p   0.01000000 odds          0.0   log odds      -4.5951
p   0.00100000 odds          0.0   log odds      -6.9068
p   0.00000003 odds          0.0   log odds     -17.3221
```

Both of these are typically used to *display* results, rather than to do computations. But **log-probabilities** are widely used for computation as well as display. They help solve *numerical problems* in probability calculations.

# Log-probabilities

The probability of multiple *independent* random variables taking on a set of values can be computed from the product: $P(X,Y,Z)=P(X)P(Y)P(Z)$ and in general:

$$P(X_1 = x_i, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i = x_i)$$

**Underflow:** multiplying values <1 leads to numerical issues: we will get floating point underflow. Instead, it is numerically more reliable to manipulate **log probabilities**, which can be summed instead of multiplied:

$$\log P(x_1, \ldots, x_n) = \sum_{i=1}^{n} \log P(x_i)$$

The **log-likelihood** is just $\log P(B \mid A)$, which is often more convenient than the raw likelihood.

# Likelihood function

- When talking about **likelihood**, we often write $L(x_i)$ to mean the likelihood of xi. The likelihood is not a probability. It is a function of data, and $L(x_i)=f_X(x_i)$
- For example, consider the empirical PMF of Romeo and Juliet. This gives the probability of seeing any given character. The likelihood of seeing all of the characters, given our per-character probability model, is:

$$\mathcal{L}(c_1, c_2, \ldots c_n) = \prod_i P(c_i)$$

# Example: Romeo & Juliet

This is the likelihood of the text.

[71]:
```
def likelihood(sequence, p):

    result = 1.0
    for char in sequence:
        # this is bad: multiplying lots of small numbers
        # will underflow very quickly
        result = result * p[char]
    return result
```

[72]:
```
print(likelihood(rj_chars, rj_pmf)) # huh?
```

```
0.0
```

# Example: Romeo & Juliet

The log-likelihood does not have this problem with underflow:

$$\log \mathcal{L}(c_1, c_2, \dots\ c_n) = \sum_i \log P(c_i)$$

[74]:
```
def log_likelihood(sequence, p):
    result = 0.0
    for char in sequence:
        ## summing logs is stable numerically
        result = result + np.log(p[char])
    return result
```

[75]:
```
#
print(log_likelihood(rj_chars, rj_pmf))
```

```
-344778.03999527503
```

# Comparing Log-Likelihoods

```
[77]: macbeth_chars = numerify("data/macbeth.txt")

      ## A positive value indicates more likely to be "like"
      # Romeo and Juliet; negative is evidence
      ## in favour of Metamorphosis (i.e. written by Kafka)
      print(log_likelihood(macbeth_chars, rj_pmf)-
            log_likelihood(macbeth_chars, metamorphosis_pmf))

      113.97323849247186
```

```
[78]: ## We could repeat this with another text, in this case
      ## "The Trial", by Kafka
      trial_chars = numerify("data/the_trial.txt")

      ## A positive value indicates more likely to be "like" Romeo and Juliet
      print(log_likelihood(trial_chars , rj_pmf)-
            log_likelihood(trial_chars, metamorphosis_pmf))

      -928.3458629352972
```

Thank you

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